

ARMA Models

AR · MA · ARMA · Estimation · Diagnostics · Forecasting

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By the end of this chapter, you will be able to:

- write an $AR(p)$, $MA(q)$, $ARMA(p, q)$ model in lag-operator form;
- state the stationarity and invertibility conditions in terms of polynomial roots;
- read the ACF/PACF *signatures* of AR, MA, ARMA;
- estimate an ARMA model by maximum likelihood;
- validate the model with residual diagnostics (Ljung–Box, normality);
- produce h -step-ahead forecasts with confidence intervals.

Chapter outline

1. Autoregressive process $AR(p)$
2. Moving-average process $MA(q)$
3. $ARMA(p, q)$
4. Box–Jenkins identification (ACF / PACF signatures)
5. Estimation: MLE, conditional sum of squares
6. Model selection: AIC / BIC
7. Residual diagnostics
8. Forecasting: point forecast and confidence interval

Autoregressive process $AR(p)$

1. Autoregressive process $AR(p)$

Definition

An autoregressive process of order p satisfies

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + \varepsilon_t, \quad \varepsilon_t \sim WN(0, \sigma^2).$$

The current value depends on a finite history of *past values* of the series, plus an innovation ε_t .

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Special case $AR(1)$:

$$X_t = \phi X_{t-1} + \varepsilon_t.$$

- ϕ controls persistence.
- Large $|\phi| \Rightarrow$ slow mean reversion.

AR(p) in lag-operator form

Recall $LX_t = X_{t-1}$ and define the AR polynomial

$$\Phi(z) = 1 - \phi_1 z - \phi_2 z^2 - \dots - \phi_p z^p.$$

The AR(p) writes compactly as

$$\Phi(L) X_t = \varepsilon_t.$$

Stationarity condition

AR(p) admits a (unique, causal) stationary solution if and only if all roots of $\Phi(z) = 0$ lie *strictly outside* the unit disc, i.e. $|z_i| > 1$ for $i = 1, \dots, p$.

AR(1): root is $z = 1/\phi$, condition becomes $|\phi| < 1$.

AR(1): moments and ACF

Assume $|\phi| < 1$. Iterating $X_t = \phi X_{t-1} + \varepsilon_t$ backward gives

$$X_t = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j} \quad (\text{MA}(\infty) \text{ representation}).$$

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Hence

$$\mathbb{E}[X_t] = 0, \quad \text{Var}(X_t) = \frac{\sigma^2}{1 - \phi^2}, \quad \rho(h) = \phi^{|h|}.$$

- ACF **decays geometrically** with rate ϕ .
- PACF **cuts off** after lag 1: $\alpha(1) = \phi$, $\alpha(h) = 0$ for $h \geq 2$.

AR(2): stationarity triangle

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t.$$

Stationarity $\Leftrightarrow (\phi_1, \phi_2)$ in the triangle

$$\phi_1 + \phi_2 < 1, \quad \phi_2 - \phi_1 < 1, \quad -1 < \phi_2 < 1.$$

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Yule–Walker equations give the ACF recursively:

$$\rho(h) = \phi_1 \rho(h-1) + \phi_2 \rho(h-2), \quad h \geq 1.$$

with $\rho(0) = 1$, $\rho(1) = \phi_1/(1 - \phi_2)$.

Moving-average process $MA(q)$

2. Moving-average process MA(q)

Definition

$$X_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}, \quad \varepsilon_t \sim WN(0, \sigma^2).$$

The current value is a finite linear combination of *present and past shocks*.

2. Moving-average process MA(q)

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Compact form with $\Theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$:

$$X_t = \Theta(L) \varepsilon_t.$$

Every MA(q) is stationary (finite linear combination of white noise).

MA(q): moments and ACF

$$\mathbb{E}[X_t] = 0, \quad \text{Var}(X_t) = \sigma^2(1 + \theta_1^2 + \cdots + \theta_q^2).$$

Autocovariance:

$$\gamma(h) = \begin{cases} \sigma^2(\theta_h + \theta_1\theta_{h+1} + \cdots + \theta_{q-h}\theta_q) & 1 \leq h \leq q, \\ 0 & h > q. \end{cases}$$

MA(q): moments and ACF

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- ACF **cuts off** at lag q – the defining MA signature.
- PACF decays geometrically (no cutoff).

Invertibility of MA(q)

An MA(q) is *invertible* if it can be rewritten as an AR(∞):

$$\varepsilon_t = X_t - \pi_1 X_{t-1} - \pi_2 X_{t-2} - \dots$$

with $\sum |\pi_j| < \infty$.

Invertibility condition

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MA(1) example: $X_t = \varepsilon_t + \theta \varepsilon_{t-1}$ is invertible iff $|\theta| < 1$. Without invertibility, two parameter sets give the same ACF – the model is not identifiable.

ARMA(p, q)

3. ARMA(p, q)

Definition

$$\Phi(L) X_t = \Theta(L) \varepsilon_t,$$

or, expanded,

$$X_t = \sum_{i=1}^p \phi_i X_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j}.$$

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- AR part captures persistence in past values.
- MA part captures persistence of past shocks.
- Special cases: $\text{ARMA}(p, 0) = \text{AR}(p)$, $\text{ARMA}(0, q) = \text{MA}(q)$.

ARMA(p, q): existence and identifiability

Conditions

- **Stationarity:** roots of $\Phi(z) = 0$ outside the unit disc.
- **Invertibility:** roots of $\Theta(z) = 0$ outside the unit disc.
- **Identifiability:** Φ and Θ have no common root (no AR/MA cancellation).

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Under these three conditions, the ARMA process is a stationary causal invertible solution of the equation, with absolutely summable MA(∞) representation $X_t = \Phi^{-1}(L)\Theta(L)\varepsilon_t$.

Box–Jenkins identification

4. ACF / PACF signatures

Model	ACF $\rho(h)$	PACF $\alpha(h)$
AR(p)	exponential decay	cuts off after lag p
MA(q)	cuts off after lag q	exponential decay
ARMA(p, q)	exponential decay (after lag q)	exponential decay (after lag p)

4. ACF / PACF signatures

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Box–Jenkins identification rule

- PACF cuts off at lag p & ACF decays $\Rightarrow$ AR(p).
- ACF cuts off at lag q & PACF decays $\Rightarrow$ MA(q).
- Both decay $\Rightarrow$ ARMA – compare AIC/BIC across orders.

Bartlett bands $\pm 1.96/\sqrt{T}$ delimit “zero”.

- “Cuts off after lag k ” means $\hat{\rho}(h)$ (or $\hat{\alpha}(h)$) drops inside the band for all $h > k$.
- “Decays exponentially” means $\hat{\rho}(h)$ stays outside the band for many lags but shrinks.

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Caveat: on real data, the picture is often noisy. Identification gives *candidate* orders – the final choice uses information criteria and residual diagnostics.

Estimation

5. Estimation – Yule–Walker for AR(p)

Multiplying $X_t = \sum \phi_i X_{t-i} + \varepsilon_t$ by X_{t-h} and taking expectations gives the Yule–Walker system:

$$\rho(h) = \phi_1 \rho(h-1) + \cdots + \phi_p \rho(h-p), \quad h = 1, \dots, p.$$

In matrix form $\mathbf{R}\phi = \mathbf{r}$:

$$\begin{pmatrix} 1 & \rho(1) & \cdots & \rho(p-1) \\ \rho(1) & 1 & \cdots & \rho(p-2) \\ \vdots & & \ddots & \vdots \\ \rho(p-1) & \cdots & \rho(1) & 1 \end{pmatrix} \begin{pmatrix} \phi_1 \\ \vdots \\ \phi_p \end{pmatrix} = \begin{pmatrix} \rho(1) \\ \vdots \\ \rho(p) \end{pmatrix}.$$

Plug $\hat{\rho}$ in place of ρ to obtain $\hat{\phi}_{YW}$. Method-of-moments style; consistent but not efficient.

Conditioning on the first p observations and assuming $\varepsilon_t = 0$ for $t \leq 0$:

$$\hat{\theta}_{\text{CSS}} = \arg \min_{\theta} \sum_{t=p+1}^T \hat{\varepsilon}_t(\theta)^2,$$

with $\hat{\varepsilon}_t$ obtained recursively from the model equation. Equivalent to MLE under Gaussian innovations and large T , easier to compute.

Under $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$, the joint likelihood of $(X_1, \dots, X_T)$ is multivariate normal with covariance $\Gamma_T(\boldsymbol{\theta})$:

$$\log \mathcal{L}(\boldsymbol{\theta}, \sigma^2) = -\frac{T}{2} \log(2\pi) - \frac{1}{2} \log |\Gamma_T| - \frac{1}{2} \mathbf{X}^\top \Gamma_T^{-1} \mathbf{X}.$$

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Maximised numerically (state-space + Kalman filter; the standard backend of `statsmodels.SARIMAX`). Provides asymptotic standard errors and t -statistics for each coefficient.

Estimation – Python (statsmodels)

```
from statsmodels.tsa.arima.model import ARIMA

model = ARIMA(y, order=(2, 0, 1))    # ARMA(2,1)
fit    = model.fit()
print(fit.summary())
print("AIC =", fit.aic, " BIC =", fit.bic)
print("Coefs:", fit.params)
```

- `order=(p, d, q)` – here $d = 0$ for ARMA.
- `fit.summary()` prints estimates, standard errors, z-statistics, p -values, AIC, BIC, Ljung–Box.
- `fit.resid` gives the residuals; use them in diagnostics.

Model selection

6. Information criteria

Compare candidate orders by penalised log-likelihood:

$$\text{AIC} = -2 \log \mathcal{L} + 2k, \quad \text{BIC} = -2 \log \mathcal{L} + k \log T,$$

where $k = p + q + 1$ is the number of free parameters.

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- AIC favours predictive accuracy; can over-fit on small T .
- BIC penalises complexity more (consistent under the true order assumption).

Practical rule: when AIC and BIC disagree, prefer the parsimonious BIC choice and check it with residual diagnostics.

Residual diagnostics

7. Residual diagnostics

Once $\hat{\theta}$ is obtained, compute the residuals $\hat{\varepsilon}_t = X_t - \hat{X}_{t|t-1}$. A well-fitted model satisfies:

- $\hat{\varepsilon}_t$ should be **white noise**;
- residual ACF inside the Bartlett band;
- Ljung–Box test does not reject H_0 (white noise);
- residuals approximately normal (QQ-plot, Jarque–Bera).

7. Residual diagnostics

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- residuals approximately normal (QQ-plot, Jarque–Bera).

Failure of any test $\Rightarrow$ revisit identification: try a larger AR/MA order, or look for a structural break / volatility effect (Chapter 6).

Ljung–Box test on residuals

Test

$$Q_{\text{LB}}(m) = T(T+2) \sum_{h=1}^m \frac{\hat{\rho}_{\hat{\epsilon}}(h)^2}{T-h}.$$

Under H_0 (residuals are WN), and after fitting an ARMA(p, q),

$$Q_{\text{LB}}(m) \xrightarrow{d} \chi_{m-p-q}^2 \quad (\text{degrees of freedom adjusted for fitted parameters}).$$

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Convention. Test at $m = 12, 24$ for monthly data; $m = 10, 20$ for short series. Reject if $p\text{-value} < 0.05$.

Forecasting

8. Forecasting an ARMA(p, q)

Best linear forecast at horizon h :

$$\hat{X}_{T+h|T} = \mathbb{E}[X_{T+h} \mid X_T, X_{T-1}, \dots].$$

For AR(1): $\hat{X}_{T+h|T} = \phi^h X_T \rightarrow 0$ as $h \rightarrow \infty$.

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For AR(1): $\hat{X}_{T+h|T} = \phi^h X_T \rightarrow 0$ as $h \rightarrow \infty$. Forecast error variance grows with h :

$$\text{Var}(X_{T+h} - \hat{X}_{T+h|T}) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2,$$

where ψ_j are the MA(∞) coefficients ($X_t = \sum \psi_j \varepsilon_{t-j}$).

Forecast confidence intervals

Under Gaussian innovations, a 95% confidence interval for X_{T+h} is

$$\hat{X}_{T+h|T} \pm 1.96 \sigma \sqrt{\sum_{j=0}^{h-1} \psi_j^2}.$$

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- For stationary ARMA, the CI width tends to a finite limit (standard deviation of the unconditional distribution).
- For ARIMA (with $d \geq 1$), the CI grows without bound – consistent with the unit root.

Forecasting – Python (statsmodels)

```
fit = ARIMA(y, order=(2, 0, 1)).fit()

fc      = fit.get_forecast(steps=12)
mean    = fc.predicted_mean
ci_low  = fc.conf_int(alpha=0.05).iloc[:, 0]
ci_high = fc.conf_int(alpha=0.05).iloc[:, 1]

import matplotlib.pyplot as plt
plt.plot(y, label="observed")
plt.plot(range(len(y), len(y) + 12), mean, "r-", label="forecast")
plt.fill_between(range(len(y), len(y) + 12), ci_low, ci_high,
                 color="red", alpha=0.2)
plt.legend(); plt.show()
```

Box-Jenkins workflow

The Box–Jenkins workflow

1. **Stationarise** (Chapter 4): differencing, log-transform if needed.
2. **Identify** candidate (p, q) from ACF / PACF signatures.
3. **Estimate** each candidate by MLE.
4. **Select** the model with lowest AIC / BIC, subject to parsimony.
5. **Diagnose** the residuals (Ljung–Box, QQ-plot).
6. **Forecast** with confidence bands.

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If **diagnostics fail**, return to step 2 with a richer order or consider seasonal effects (SARIMA) or volatility (Chapter 6).

Take-home messages

- ARMA combines past values (AR) and past shocks (MA).
- Stationarity = AR roots outside the unit disc; invertibility = MA roots outside the unit disc.
- ACF / PACF tell you the order: *cutoff* reveals AR or MA; *double decay* suggests ARMA.
- Estimation: MLE in `statsmodels`; selection by AIC / BIC; residuals must be white noise.
- Forecast intervals widen with horizon at a rate dictated by the $MA(\infty)$ representation.